Single Page Explanation

What this project is, and why it works — in plain English.

See also: README · Technical Project Document · Software Controller Solution · Setup

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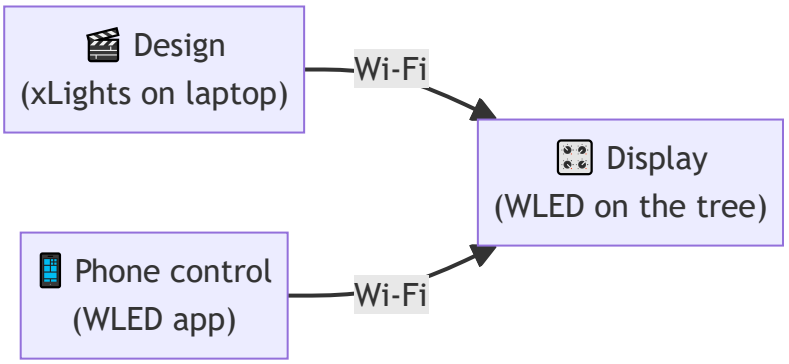
The big idea

We are building a **5–6 metre cone-shaped Christmas tree** out of **12 vertical strips of programmable LEDs**. Instead of treating those 12 strips as 12 separate lines, the system treats them as **one single 3D shape** — so an effect like a spinning sphere, a spiral, or falling snow flows smoothly across the whole tree.

Standard, off-the-shelf Christmas light controllers only come with a handful of basic patterns (flashing, fading) that **can't adapt to the physical shape of your tree**. Because we want precise control over speeds, pauses, triggers and custom animations across a 3D cone, we use two free, professional-grade tools together: **xLights** and **WLED**.

A helpful way to picture it:

> **xLights is the director's computer. WLED is the stage manager standing next to the lights.**



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xLights in plain English

xLights is **free software you install on a laptop**. Think of it like iMovie or GarageBand — but for lights instead of video or music.

- **It understands 3D shapes.** A tree is a cone, not a flat screen. If you ran a "circle" effect on a cone with normal software it would look squashed at the top. xLights builds a *virtual 3D model* of your 12-strip cone and does the maths so effects map perfectly onto the real shape.

- **It has a timeline editor.** You drag and drop effects (waves, fireworks, gradients, scrolling text) onto a timeline and stretch or shrink them to control speed, repeats and transitions.
- **It does the heavy lifting.** Your laptop is far more powerful than the tiny chip on the tree, so it does all the rendering and turns your ideas into data ready to send.

WLED in plain English

WLED is **free firmware** (software) that runs on the small ESP32 computer chip permanently attached to the tree. It is the local brain on the tree itself.

- **It speaks both "network" and "electrical".** Your laptop talks over Wi-Fi; the LED strips need fast electrical pulses. WLED catches the wireless data and instantly translates it into the signals the strips understand.
- **It has a brilliant phone app.** Connect your phone to the tree's own Wi-Fi and you get a clean screen to turn the tree on/off, change brightness, or adjust speed instantly.
- **It runs standalone.** You don't want a laptop sitting out in the cold all winter. You *upload* your finished xLights shows into the controller's memory once — after that the tree runs entirely on its own and you control everything from your phone.

Why use them together

Each tool covers the other's weakness. Together they give you unlimited design power *and* effortless everyday control.

What you need	WLED alone	xLights alone	Why the combination wins
Complex 3D matrix effects	❌ Basic 1D patterns only	✅ Flawless 3D maths	xLights **designs** it; WLED **displays** it
Wireless phone control	✅ Excellent app	❌ Needs a laptop	WLED handles **daily** phone control
Standalone operation	✅ Saves to internal flash	❌ Must stay tethered	xLights **exports** the show; WLED **plays it back**

In short: design on the laptop with xLights when you're being creative; enjoy effortless, lightweight control from your phone with WLED the rest of the winter.